```
# What is matplotlib?
# Matplotlib is a Python library used for creating static, animated, and interactive plots.

# Data visualization: It's the graphical representation of data to identify patterns, trends,
```

0 Install Matplotlib

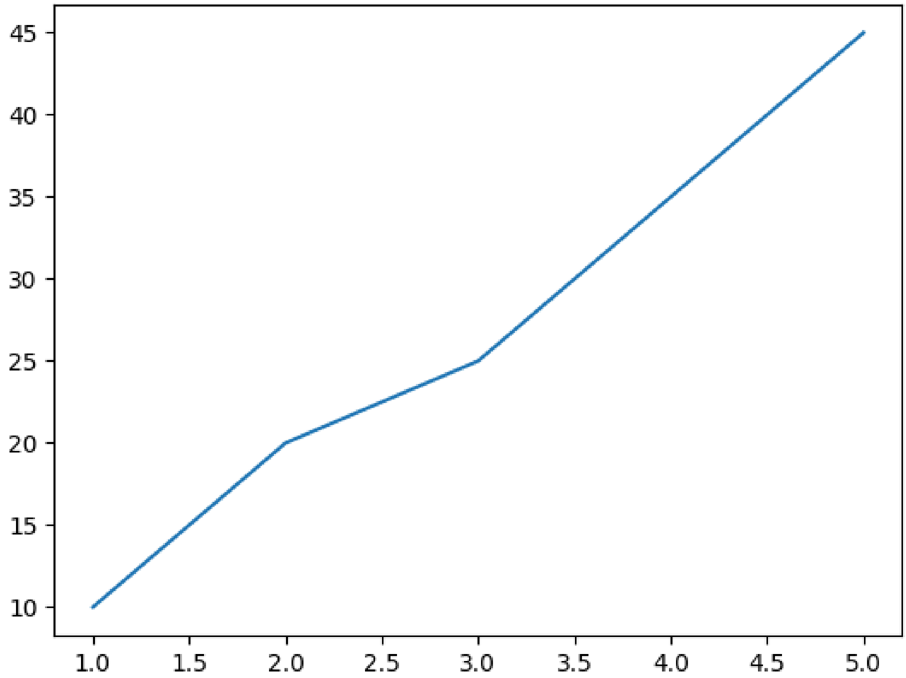
```
!pip install matplotlib
```

```
Requirement already satisfied: matplotlib in c:\users\hp\appdata\local\programs\python\python
Requirement already satisfied: contourpy>=1.0.1 in c:\users\hp\appdata\local\programs\python\
Requirement already satisfied: cycler>=0.10 in c:\users\hp\appdata\local\programs\python\pyth
Requirement already satisfied: fonttools>=4.22.0 in c:\users\hp\appdata\local\programs\python
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\hp\appdata\local\programs\python
Requirement already satisfied: numpy>=1.23 in c:\users\hp\appdata\local\programs\python\pytho
Requirement already satisfied: packaging>=20.0 in c:\users\hp\appdata\local\programs\python\p
Requirement already satisfied: pillow>=8 in c:\users\hp\appdata\local\programs\python\python3
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\hp\appdata\local\programs\python\
Requirement already satisfied: python-dateutil>=2.7 in c:\users\hp\appdata\local\programs\pyt
Requirement already satisfied: six>=1.5 in c:\users\hp\appdata\local\programs\python\python31

[notice] A new release of pip is available: 24.0 -> 25.3
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
import matplotlib.pyplot as plt

# pyplot is a collection of functions that make matplotlib work as MATLAB
```

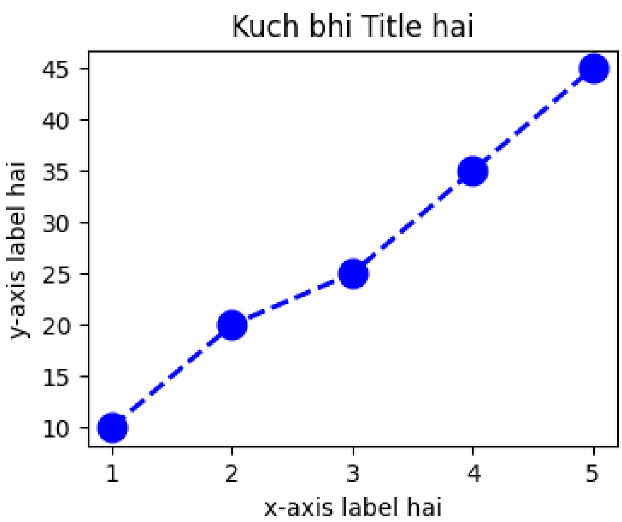
1 Create First Chart

```
# data
x = [1,2,3,4,5]
y = [10, 20, 25, 35, 45]

plt.plot(x,y) # line chart
plt.show() # display chart in jp nb
```



2 Customize Chart

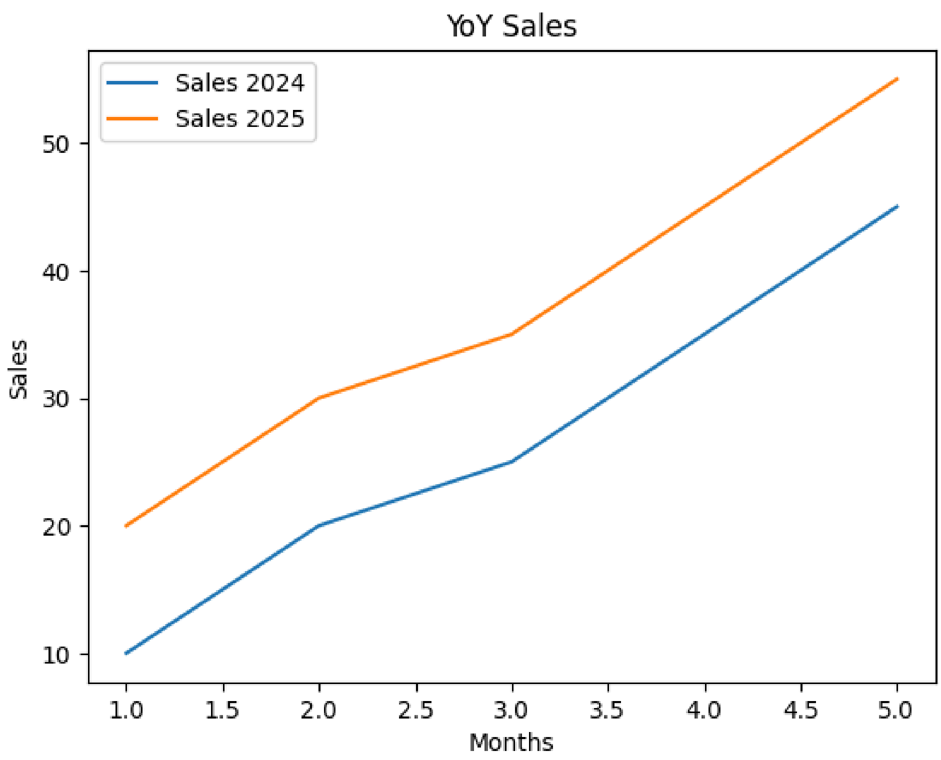
```
plt.figure(figsize=(4,3)) # figure size

plt.plot(x, y, color='blue', marker='o', linestyle='--', linewidth=2, markersize=12)

plt.title("Kuch bhi Title hai")
plt.xlabel("x-axis label hai")
plt.ylabel("y-axis label hai")

plt.show()
```



3 Advanced - Multiple Lines & Legends

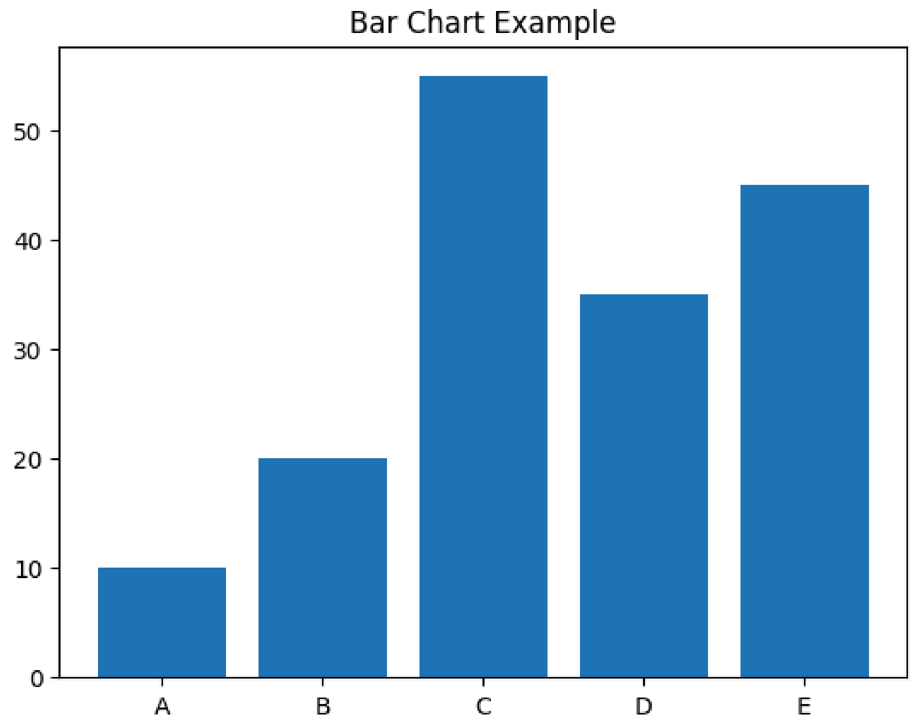
```
# data
x = [1,2,3,4,5]
y1 = [10, 20, 25, 35, 45]
y2 = [20, 30, 35, 45, 55]

plt.plot(x, y1, label='Sales 2024') # ploting y1 data
```

```
plt.plot(x, y2, label='Sales 2025') # ploting y2 data

plt.title("YoY Sales")
plt.xlabel("Months")
plt.ylabel("Sales")

plt.legend() # show legends
plt.show()
```



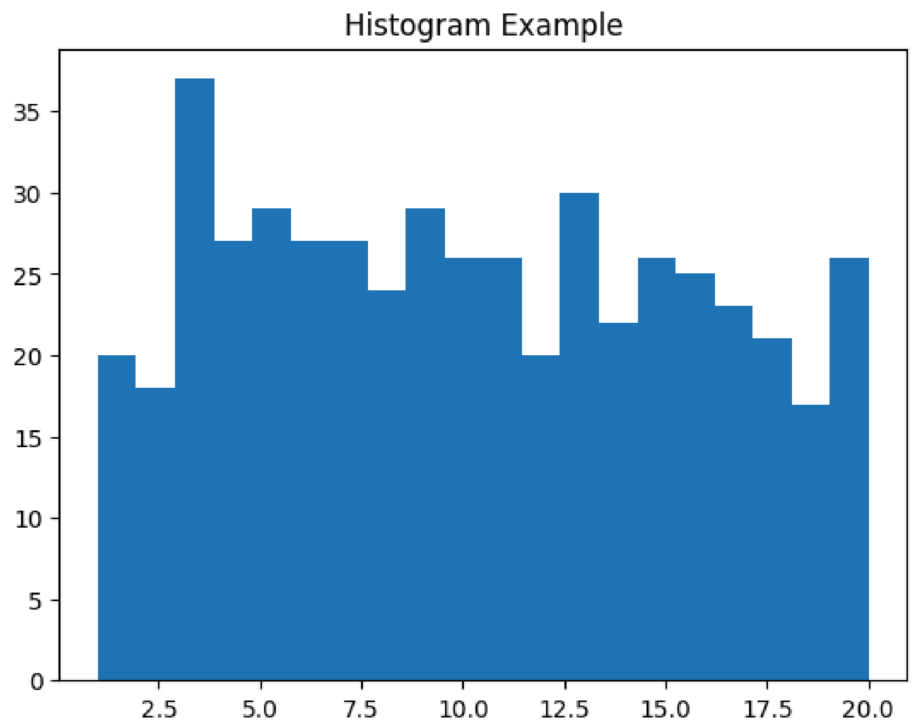
4 Bar Chart

```
# data
x = ['A','B','C','D','E']
y = [10, 20, 55, 35, 45]

plt.bar(x,y)
plt.title('Bar Chart Example')
plt.show()
```



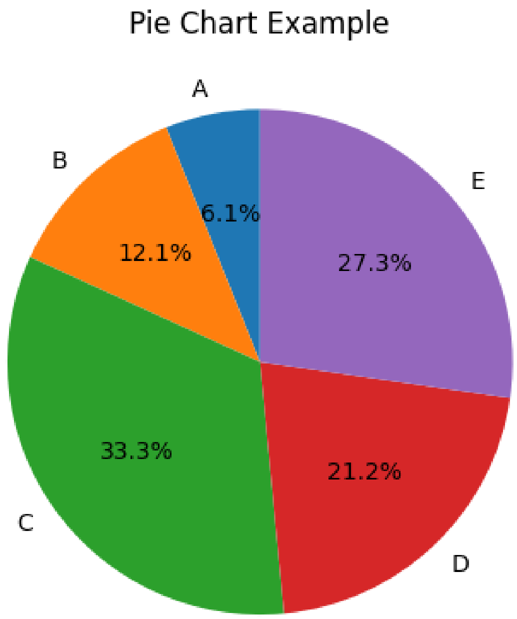
5 Histogram

```
# used for distribution analysis
# data
import random
data = [random.randint(1, 20) for _ in range(500)]

plt.hist(data, bins=20) # histogram chart
plt.title("Histogram Example")
plt.show()
```



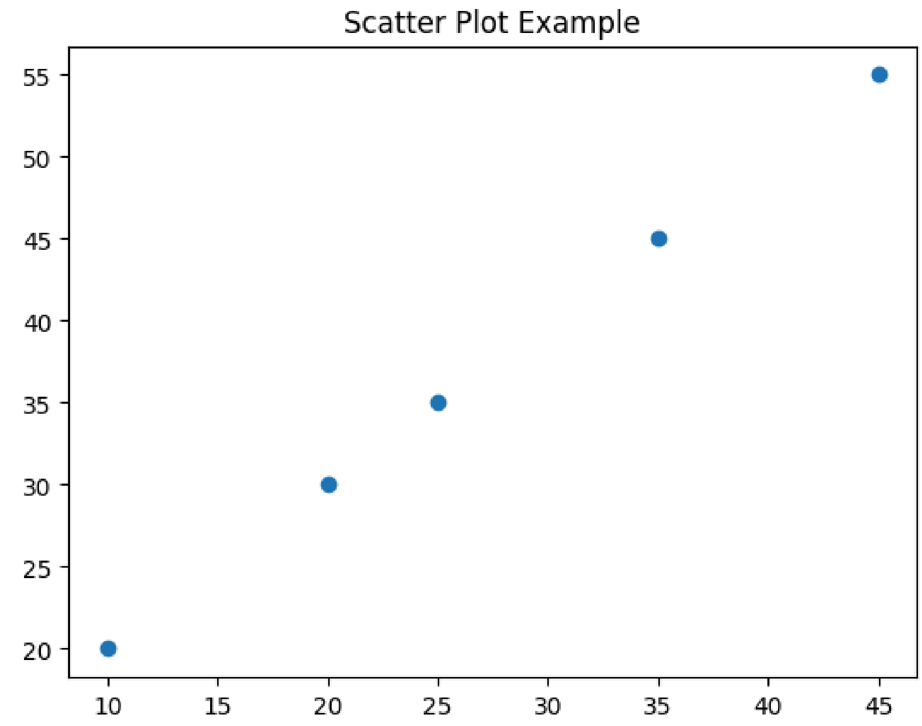
6 Pie chart

```
# used to show part-to-whole relationships/data

# data
categories = ['A','B','C','D','E']
sales = [10, 20, 55, 35, 45]

plt.pie(sales, labels=categories, autopct = '%1.1f%%', startangle=90)
plt.title("Pie Chart Example")
plt.show()
```



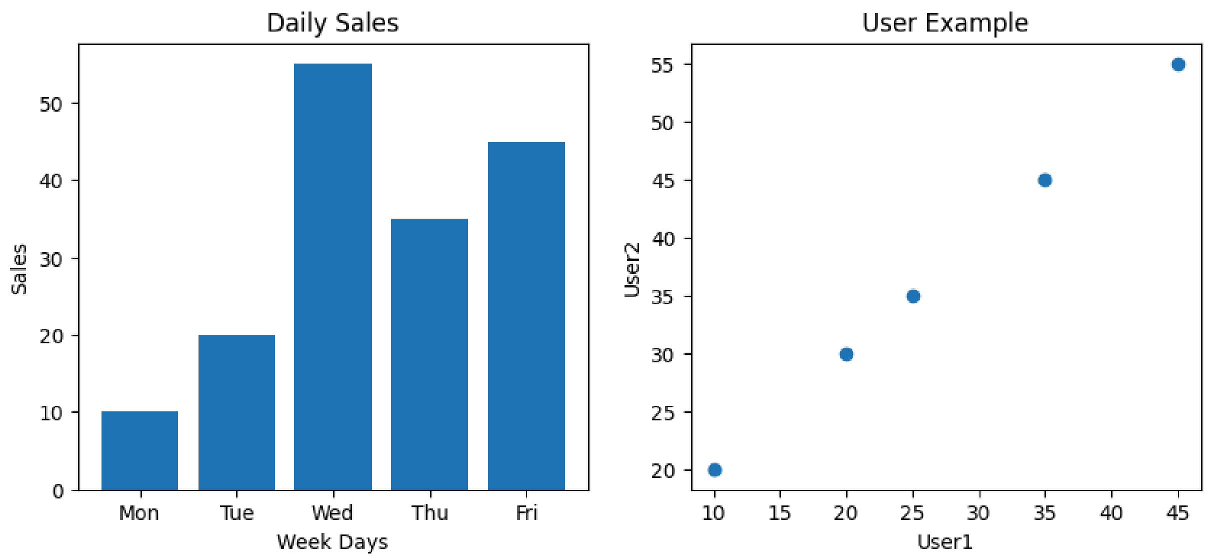
7 Scatter plot

```
# used to find relationship btwn variables

# data
y1 = [10, 20, 25, 35, 45]
y2 = [20, 30, 35, 45, 55]

plt.scatter(y1, y2)
plt.title("Scatter Plot Example")
plt.show()
```



8 Subplots

```
# used to show multiple charts in one figure

# data-1 - bar chart
categories = ['Mon','Tue','Wed','Thu','Fri']
sales = [10, 20, 55, 35, 45]

# data-2 - scatter plot
y1 = [10, 20, 25, 35, 45]
y2 = [20, 30, 35, 45, 55]

plt.figure(figsize=(10,4))

# first plot- bar chart
plt.subplot(1,2,1) # row, column, position
plt.bar(categories, sales)
plt.title("Daily Sales")
plt.xlabel("Week Days")
plt.ylabel("Sales")


# second plot- scatter chart
plt.subplot(1,2,2) # row, column, position
plt.scatter(y1, y2)
plt.title("User Example")
plt.xlabel("User1")
plt.ylabel("User2")

plt.show()
```

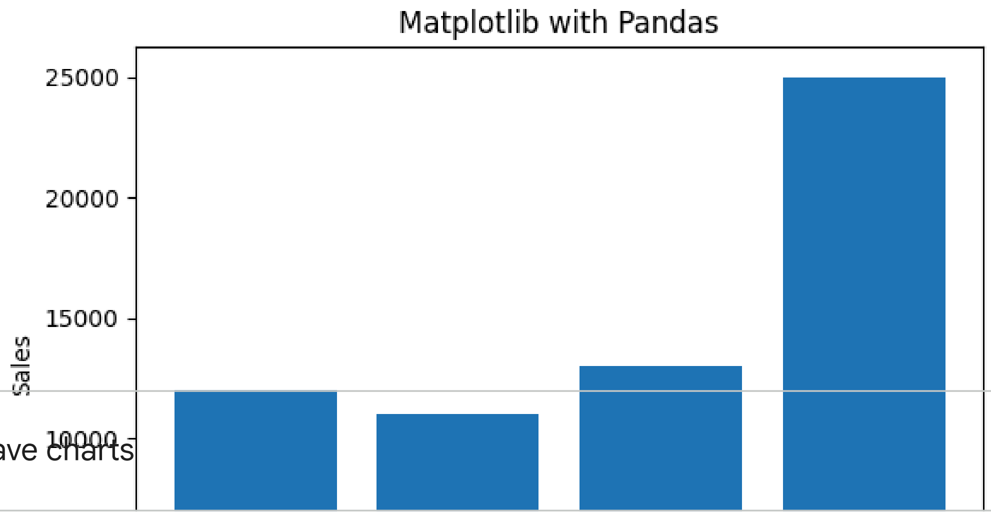


9 Matplotlib with Pandas - real data

```
# create df
import pandas as pd

# data
data = {
    'Month' : ['Jan', 'Feb', 'Mar', 'Apr'],
    'Sales' : [12000, 11000, 13000, 25000]
}

df = pd.DataFrame(data)
df
```

	Month	Sales
0	Jan	12000
1	Feb	11000
2	Mar	13000
3	Apr	25000

```
plt.bar(df['Month'], df['Sales'])
plt.title("Matplotlib with Pandas")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.show()
```



10 Save charts

```
plt.bar(df['Month'], df['Sales'])
plt.title("Matplotlib with Pandas")
plt.xlabel("Month")
plt.ylabel("Sales")
```